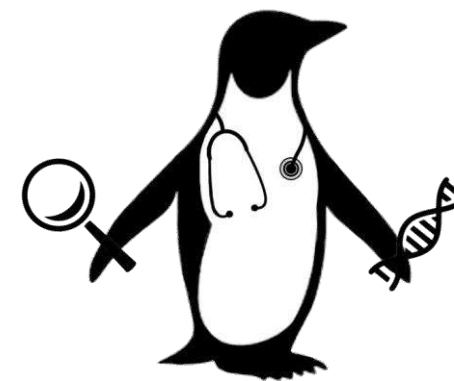




PENGQUIN

PENGQUIN:

PErsoNal Genome and QUery in healthcare and clinical practice

PhD Candidate: Elias Crum

VITO 18-month jury
Nov 20th, 2025

Advisors: Dr. Gökhan Ertaylan¹, Dr. Bart Buelens¹, Dr. Ruben Taelman², Prof. Dr. Ruben Verborgh²

1 = 

2 = 

Presentation Overview

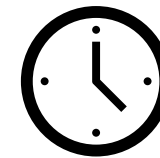
Project Review



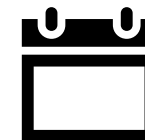
~24 Month Progress



Current Undertakings



Remaining PhD Plans

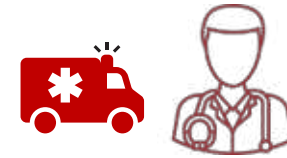


PENGQUIN:

PErsoNal Genome QUery IN
healthcare and clinical practice



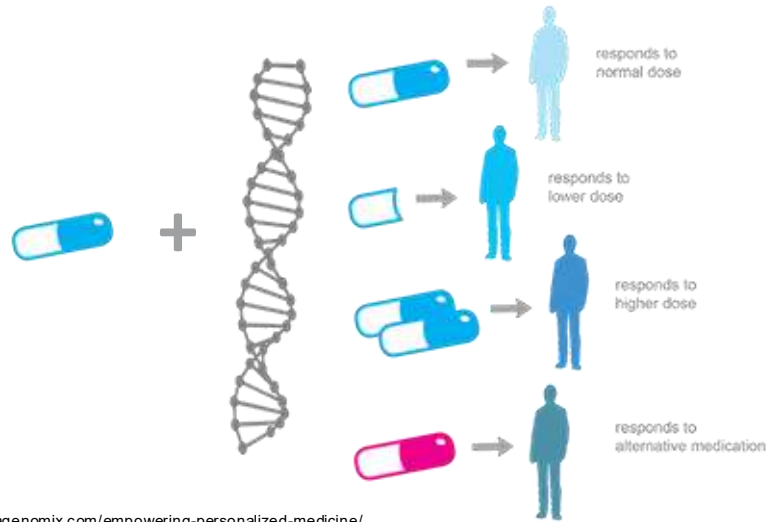
Genomic Data



Clinical Care

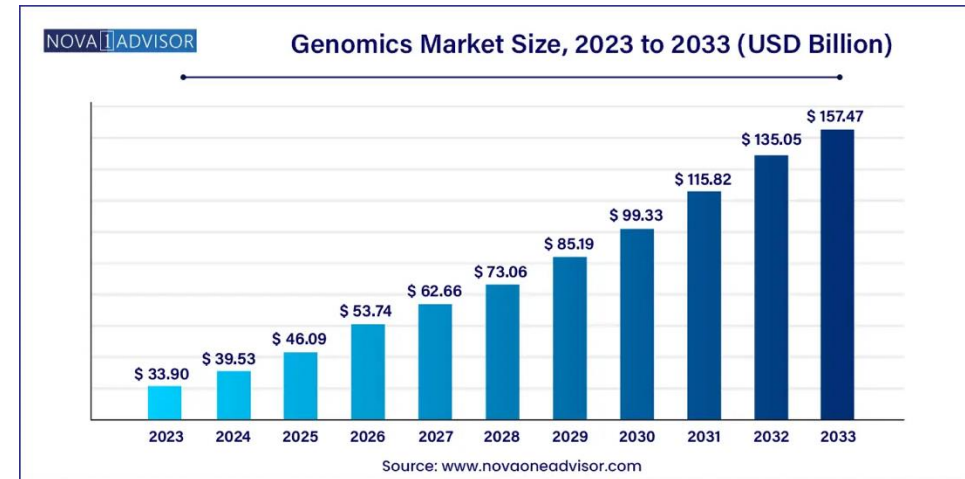
Genomics + Healthcare

A Example: Pharmacogenomics



<https://alphagenomix.com/empowering-personalized-medicine/>

B Genomics market is growing RAPIDLY



<https://www.novaoneadvisor.com/report/genomics-market>

C

Institution-centric

Data Storage



Infrastructure



?

Data Usage

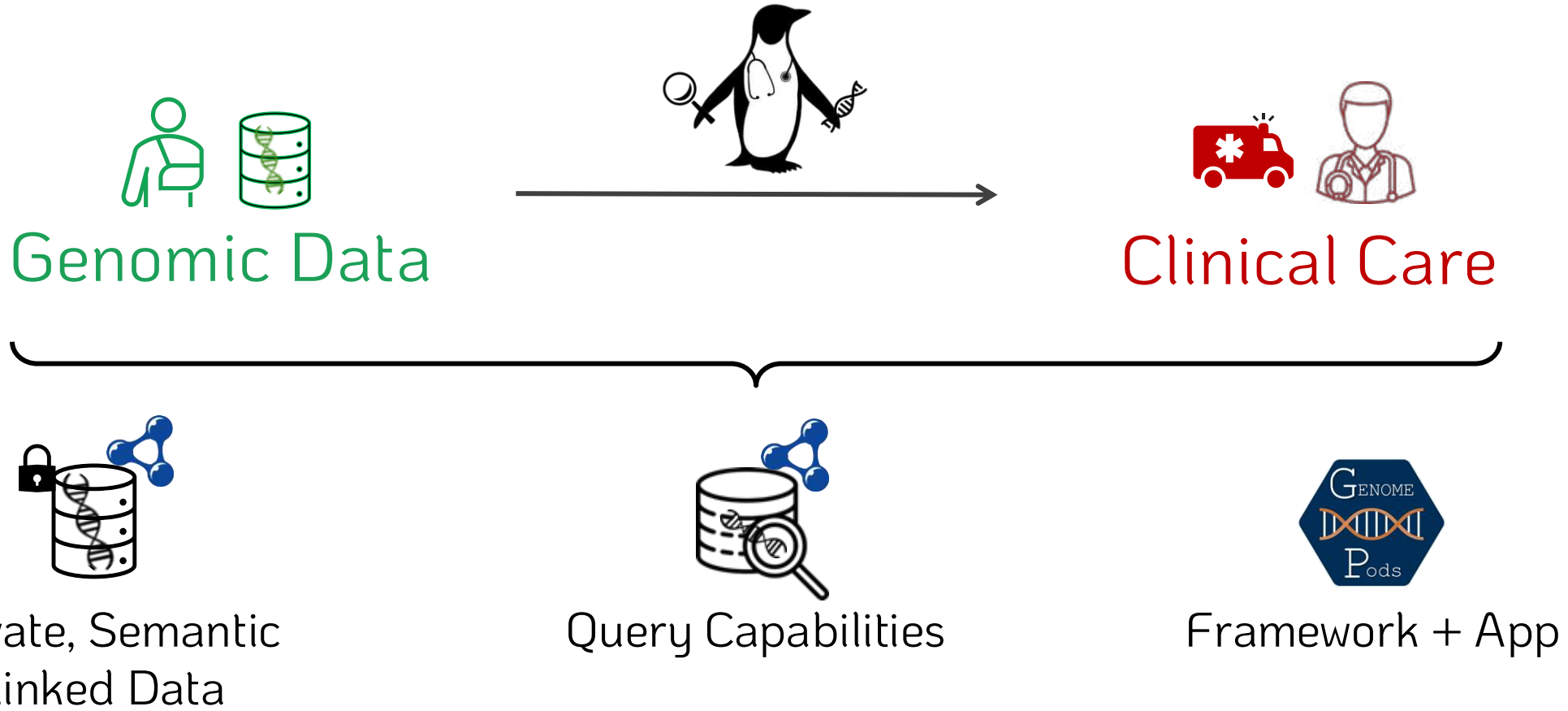


**Low Efficiency
+
High Costs**

Problem:

PENGQUIN:

PErsoNal Genome QUery IN
healthcare and clinical practice



Genomic Data for Clinical Use

-  Format is use-case specific
-  Whole Genome Sequence (WGS) data
-  Large Projects: 1+ Million Genomes (EU)¹



¹ <https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes>

Genomic Data for Clinical Use (assumption)



Format is use-case specific

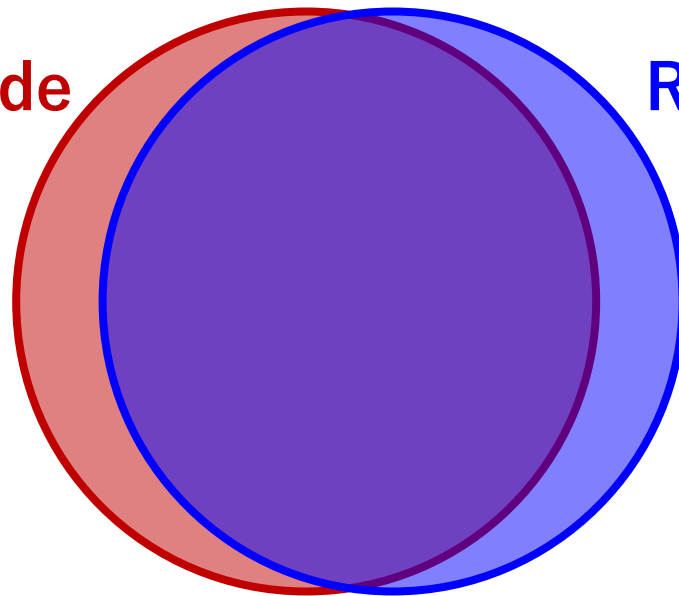


Whole Genome Sequence (WGS) data



Large Projects: 1+ Million Genomes (EU)¹

Clinical-Grade

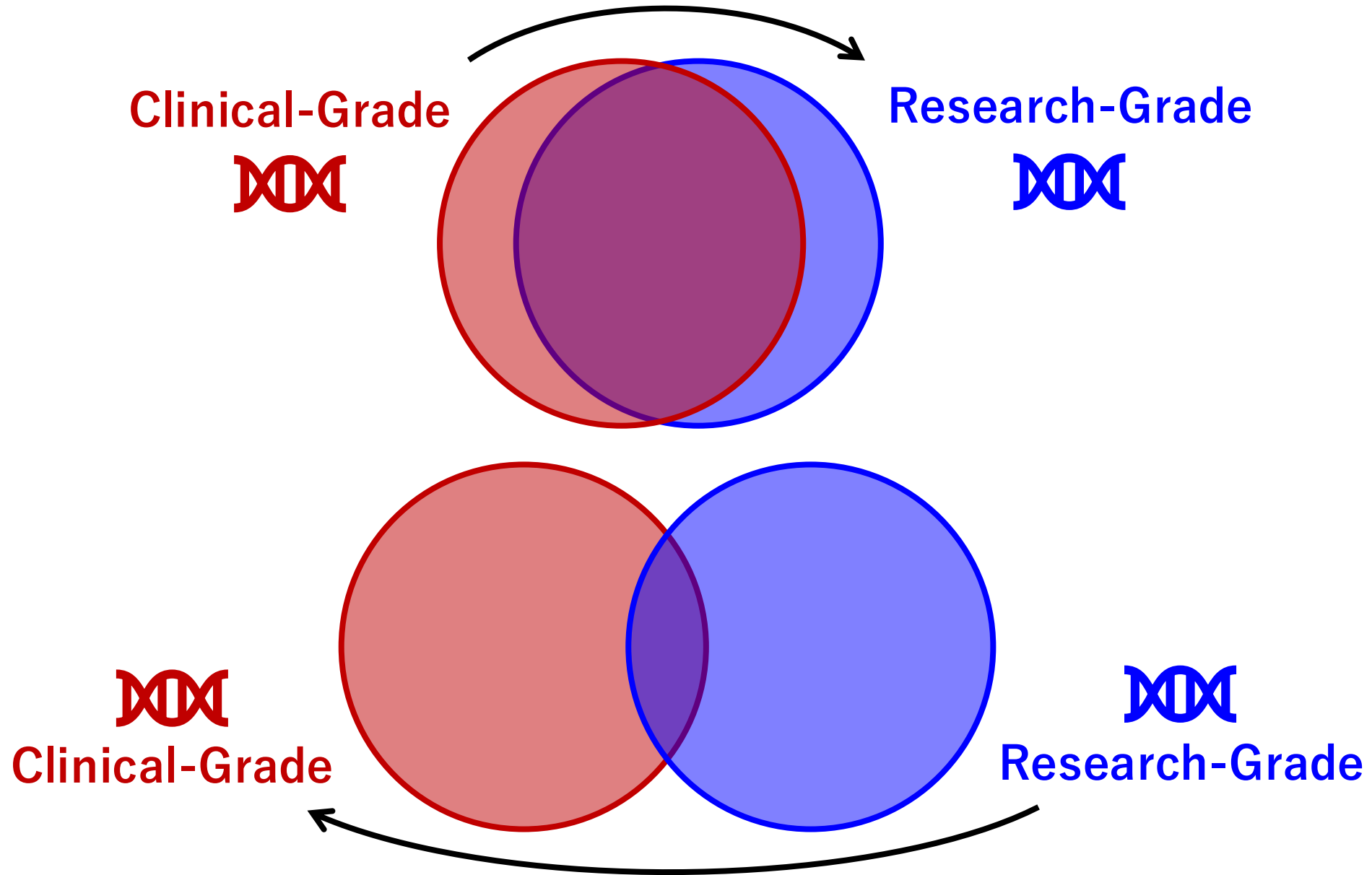


Research-Grade



¹ <https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes>

Genomic Data for Clinical Use (reality)



The current landscape

Intended use is VERY important

Clinical use $\neq$ Research use

**** Funding goes predominantly to RESEARCH-GRADE ****

Cheaper

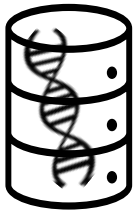
Fewer regulatory challenges

Funders / Policy makers misunderstanding

**** But how do we “translate” to clinical use?? ****

Sharing Genomic Data – Technology Considerations

Key Implementation Factors



Data Storage

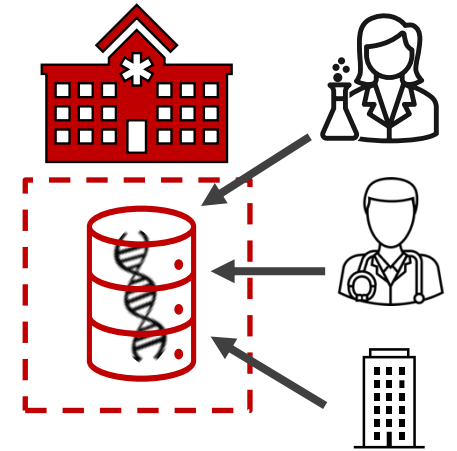
Considerations:

- patient consent
- data security
- access controls

&

Considerations:

- consent changes
- governance
- infrastructure



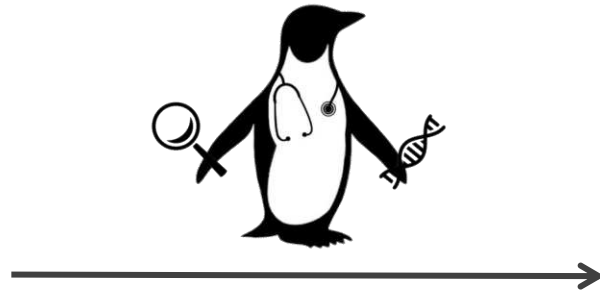
Data Sharing

A more in-depth analysis of these challenges + potential solutions detailed
in an upcoming publication in *Computers in Biology and Medicine*

PENGQUIN:

PErsoNal Genome QUery IN
healthcare and clinical practice


Genomic Data




Clinical Care



Framework + App

1. Genomic Data Interaction



Semantic
Representation

2. Semantify Genomic Data



Query Capabilities

1. Data Interaction Framework

A Web Application to:

 *Upload Data to a Solid Pod*

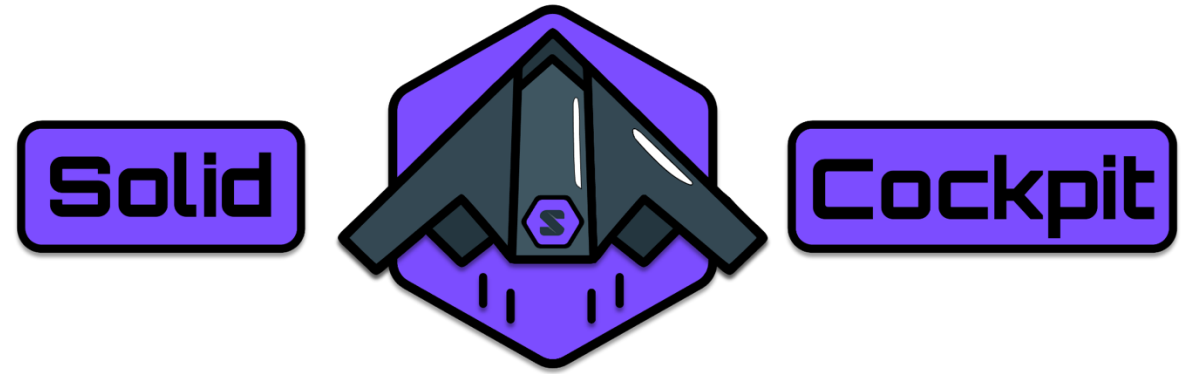
 *Browse contents of a Solid Pod*

 *Edit Privacy of (and share) data in a Solid Pod*

 *Query Data in Solid Pods + SPARQL Endpoints*

 *Includes a query cache for storing queries and results*

*** Enables patient, provider, and auditor control over data ***

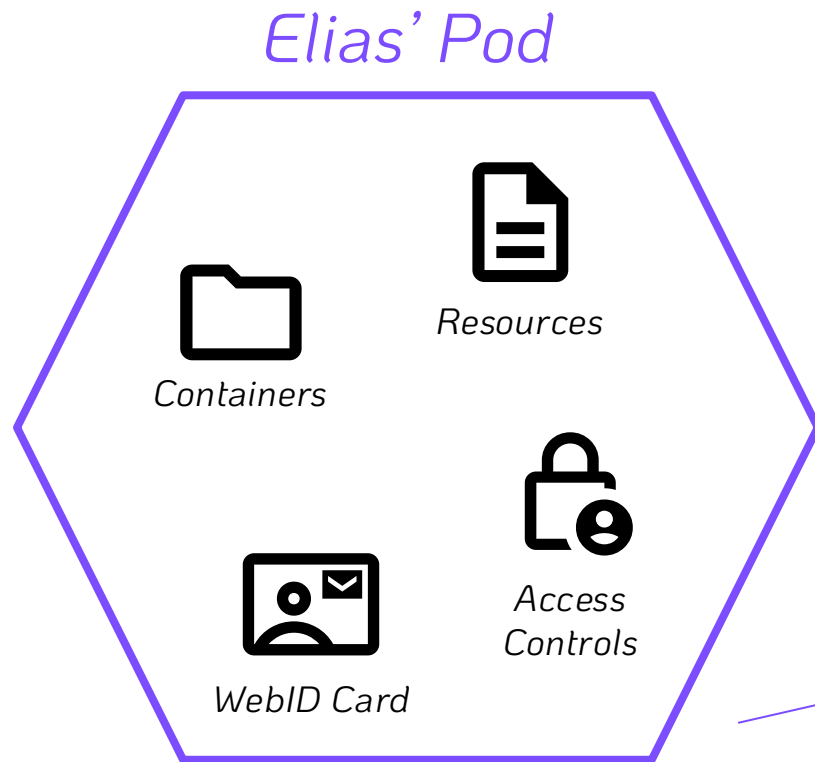


What is a Solid Pod?



Secure, Decentralized Data Storage Specification

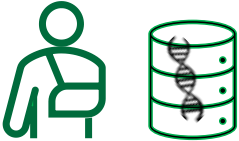
- ↳ *Pod exists on a Server (accessible to the Web)*
- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*



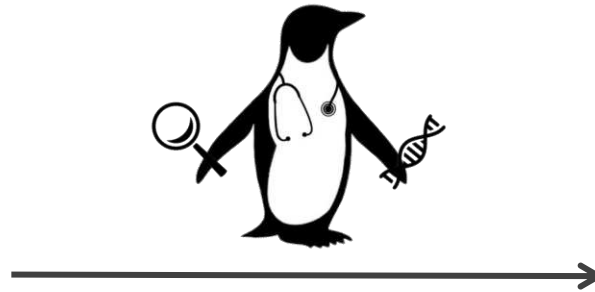
**** Enables secure, authenticated sharing via HTTP protocols ****

PENGQUIN:

PErsoNal Genome QUery IN
healthcare and clinical practice



Genomic Data

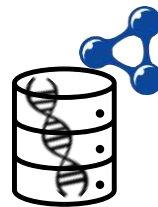


Clinical Care



Framework + App

1. Genomic Data Interaction



Semantic
Representation

2. Semantify Genomic Data



Query Capabilities

2. Genomic Data Representation

From a file-based approach

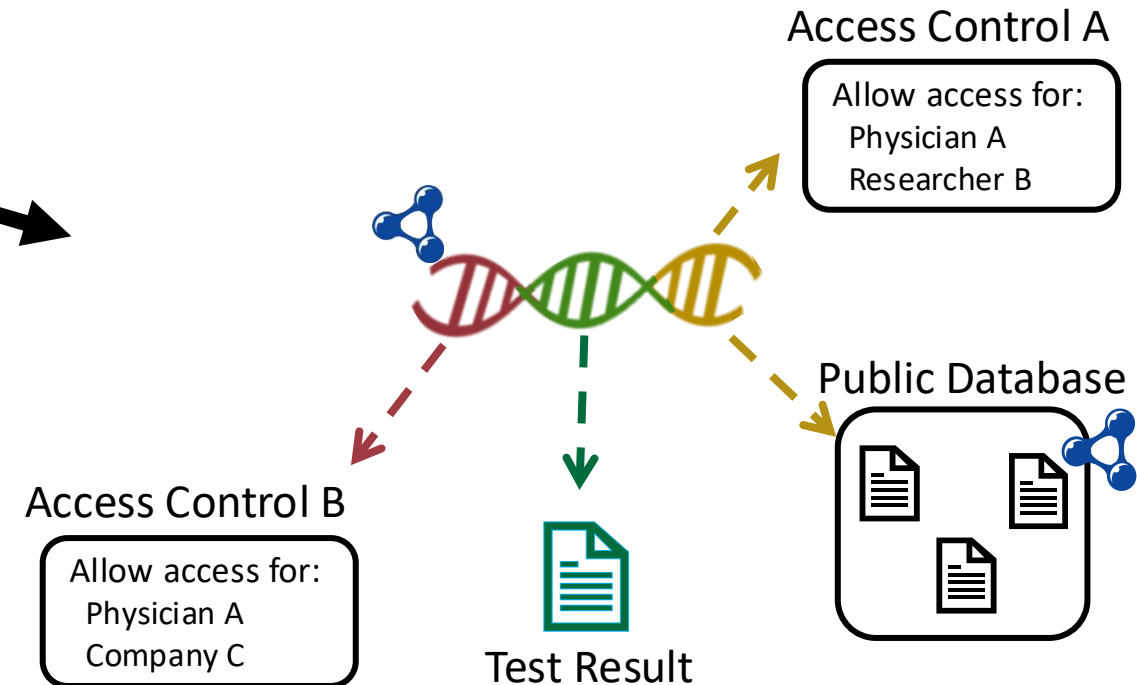


To a granular, dynamic RDF approach

Conversion using existing ontologies

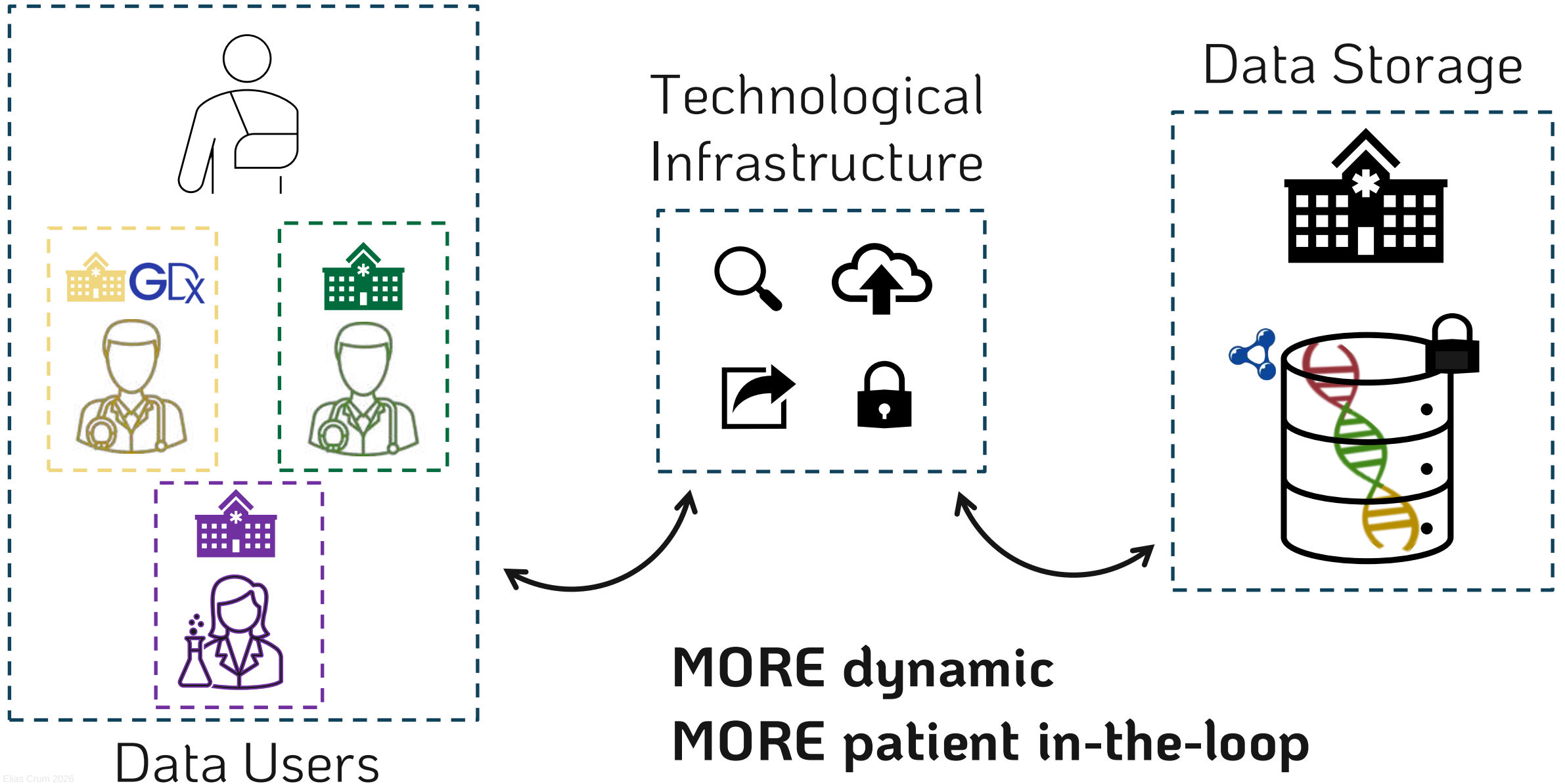
Policy definition via ODRL¹

Implementation within Solid Pod Env



¹ <https://www.w3.org/TR/odrl-model/>

Future Possibilities?



Thank you!



More PhD specifics



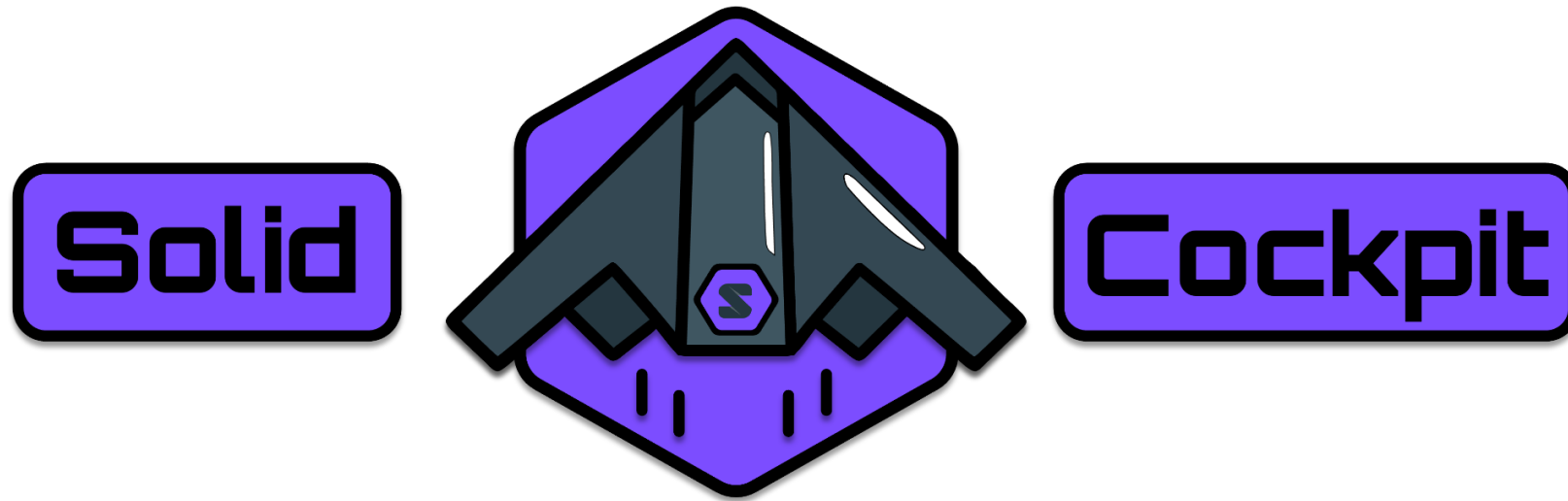
LinkedIn



GitHub

Informal Demo / Tutorial Later Today

(for those interested)



knowledgeonwebscale.github.io/solid-cockpit/

Supplemental Slides

Genomic Data

Clinical-Grade

Higher read quality
More expensive
Stricter use-regulations

Research-Grade

Lower read quality
Less expensive
Flexible use-regulations

Dimension	Clinical Grade Genome Data Sharing	Research Grade Genome Data Sharing
1. Primary Purpose and Scope	Patient care and medical decision-making.	Discovery and hypothesis-driven studies.
2. Data Accuracy and Quality	Stringent quality control (QC) and validation.	High but no absolute QC standards.
3. Regulatory Requirements	Subject to medical regulatory frameworks.	Subject to research ethics regulations.
4. Consent Requirements	Clinical-informed consent.	Research-informed consent.
5. Privacy and Confidentiality	Highest level of patient privacy required.	Strong but variable privacy measures.
6. Security Controls	Robust, enterprise-grade security.	Secure but more flexible.
7. Data Governance and Oversight	Hospital/clinic-based governance.	Institution or consortium-based governance.
8. Ethical Considerations	Focus on direct patient benefit and “do no harm.”	Focus on scientific advancement and participant welfare.
9. Infrastructure Requirements	Highly reliable, clinically validated systems.	Flexible, scalable research infrastructure.
10. Architecture Requirements	Standardized, interoperable clinical systems.	Flexible, modular architectures.
11. Legal Liabilities	High.	Moderate.
12. Interoperability	High necessity for clinical integration.	Variable.
13. Data Ownership and Control	Patient-oriented ownership with institutional custodianship.	Institution or sponsor-owned.
14. Data Sharing Scope	Limited to necessary clinical stakeholders.	Broad or consortium-based.
15. Cost and Funding Models	Healthcare reimbursement or institutional budgets.	Grant-funded or institutional research budgets.

Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review



Aim: Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

Methods: Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

Results: Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

Suggested steps forward:

Harmonize legal bases for
secondary clinical use of
research data



Mandate (GA4GH)
technical standards



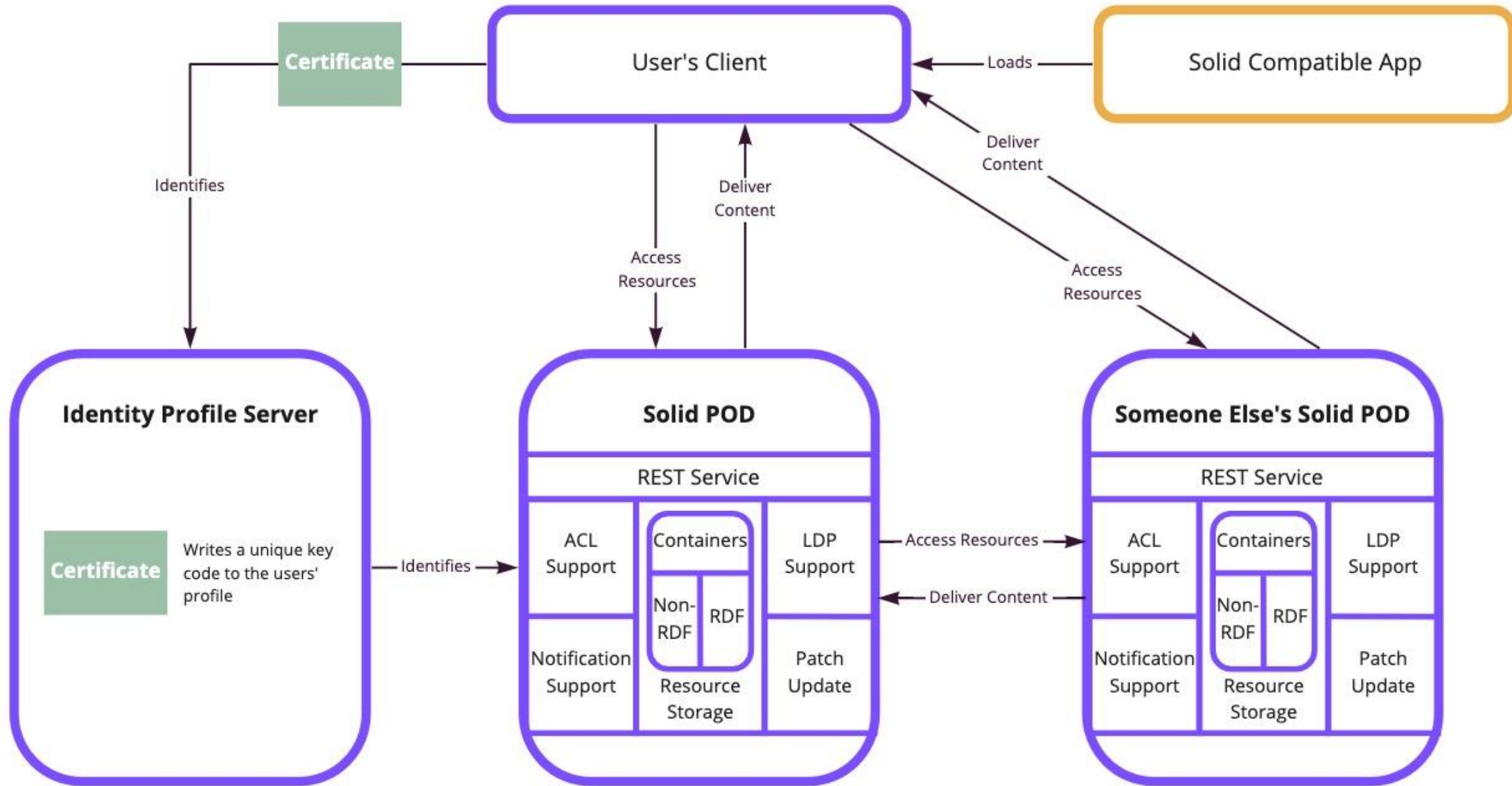
Scale laboratory
accreditation & continual
pipeline validation



Align economic incentives



Solid's Architecture



miro

Solid Pod Implementations?



BBC MyPDS App



Flemish Data Utility Company

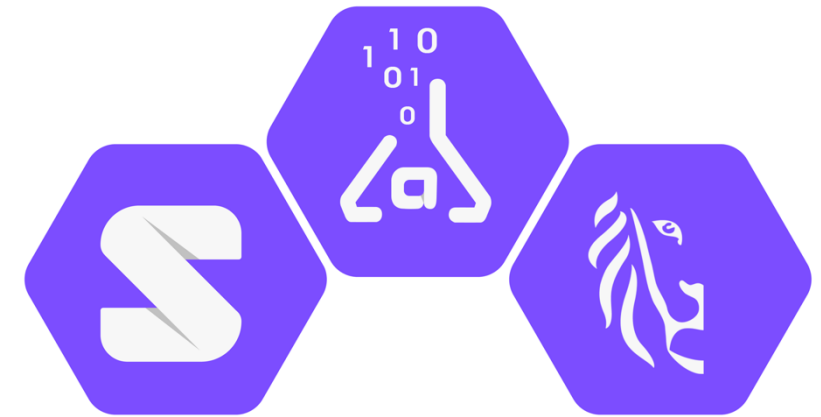


health data vaults



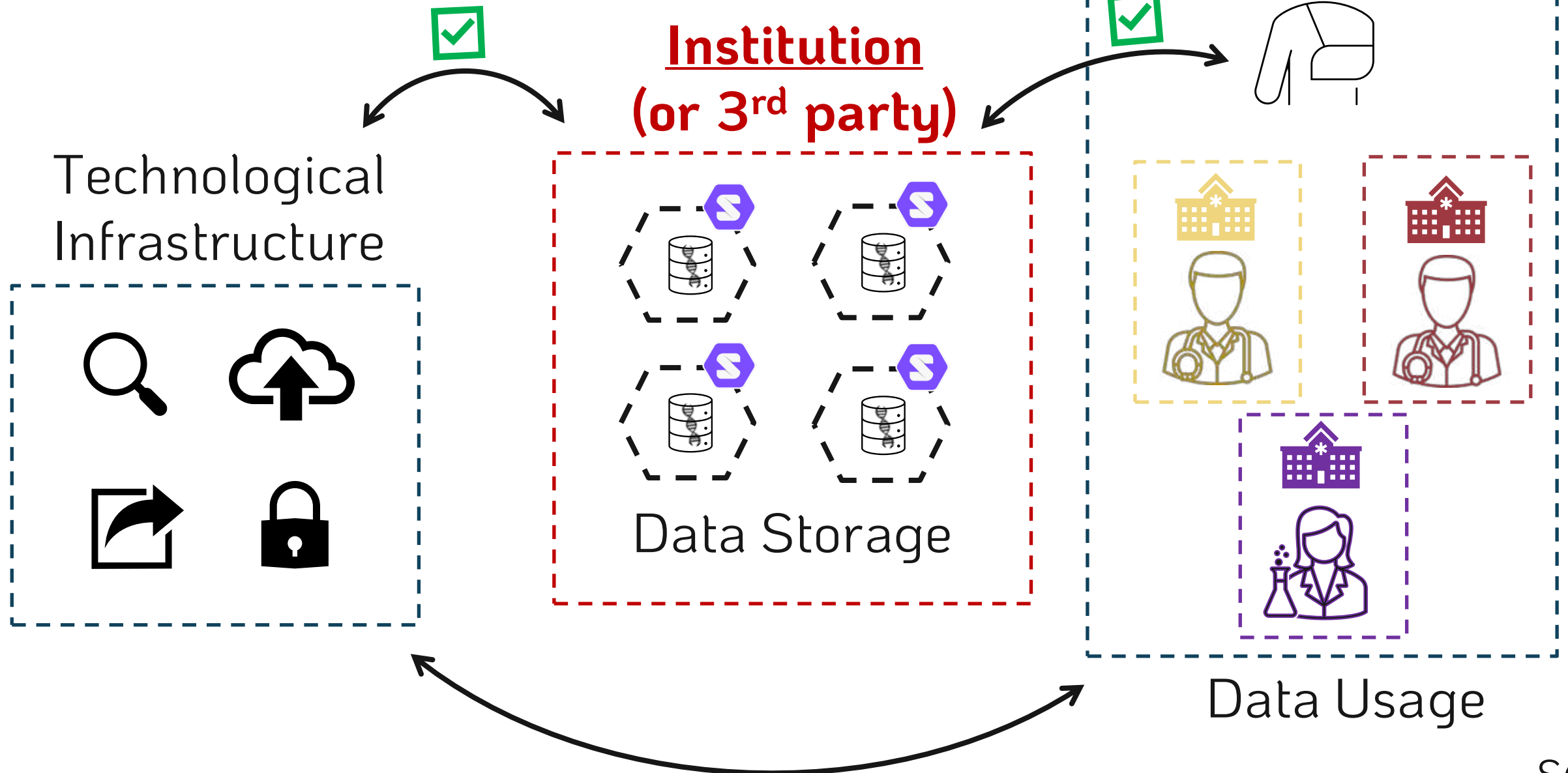
Flanders
State of the Art

Flemish Smart Dataspace

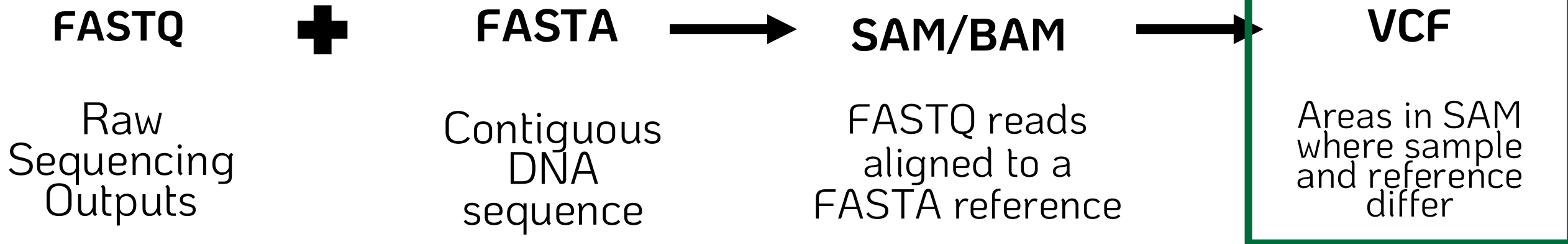


Solid Lab Use-cases

A Production Solid Solution



Semantic VCF – Sequencing Data Formats Intro



1	>gi 568336023 gb CM000663.2 Homo sapiens chromosome 1, GRCh	
2	CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	
3	ACCCTAACCCCTAACCCCTAACCCCTAACCCCAACCCCTAACCCCTAACCCCTAAC	
4	TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCT	
5	CTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	@HD VN:1.0 SO
6	CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCCTAACCC	@SQ SN:chr20
7	CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	@PG ID:TopHat
8	TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTACGCG	lign-edit-dist 2
9	TCTGACCTGAGGAGAAGCTGTGCTCCGCTTCAGAGTACCAC	20 /data/user446/m
10	CGCTCTCGCGGTGCTCTCGGGTCTGTGCTGAGGAGAACGCA	HWI-ST1145:74:C101
11	CGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCA	CCGTGTTTAAA
12	AGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCC	C BBDCCDDCCDDDD
13	CGCGCCGGCGCAGGCGCAGACACATGCTAGCGCGTCCGGGTG	AS:i:-15
14	GCGCGCCGGCGCAGGCGCAGAGACATGCTACCGCTCCAC	HWI-ST1145:74:C101
15	AGGCGCACCGCGCGGCGCAGGCGCAGAGACATGCTAGCG	TGCTGGATCAT
16	GCAGAGACGCAAGCCTACGGGCGGGGTTGGGGGGCGTGTG	G DCDDDEDDDDDD
17	GCTGGGCGGGGGGAGGGTGGCGCCGTGCACGCGCAGAACT	AS:i:-16
18	GGTAGAACCTCAGTAATCGAAAGACGGGATCGACCGCCCC	
19	GCTTGCTACGGTGCTGTGTCAGGGCGCCCTGTGCGGAC	
20	GAGTGGTGGCCAGCGCCCCCTGCTGGCGCCGGGCACTGCAG	

@FORJUSP02
CCGTCAATTC
+
AAAAAAAAAA

Q scores (as ASCII ch

Base=T, *Q*=':'=25

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
```

```
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
```

FORMAT	NA000001	NA000002	NA000003
GT:GQ:DP:HQ	0 0:48:1:51,51	1 0:48:8:51,51	1/1:43:5:.,.
GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3	0/0:41:3
GT:GQ:DP:HQ	1 2:21:6:23,27	2 1:2:0:18,2	2/2:35:4
GT:GQ:DP:HQ	0 0:54:7:56,60	0 0:48:4:51,51	0/0:61:2
GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3

```
HWI-ST1145:74:C101DACXX:7:1204:14760:4030      16      chr20      270877      50      100M      *      0      0
      GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C      DDDDDDDDDCCDDDDDDDDDEEEEEFFFFEFFFFEGHHHFGDJJIHJJIIJJIIIGGFJJIIHIIIIJJJJJJIGHHFAHGFHJHFGGHHFFDD@BB
      AS:i:-11      XM:i:2      XO:i:0      XG:i:0      MD:Z:0A85G13      NM:i:2      NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699      0      chr20      271218      50      50M4700N50M      *      0
      0      GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCCTTGGGTCTCCGAGCAGAACATCCTCAAATATGACCTCTCG
accepted hits.sam
```

What does a VCF File Look like?

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0 1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1 0:77	1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Body

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

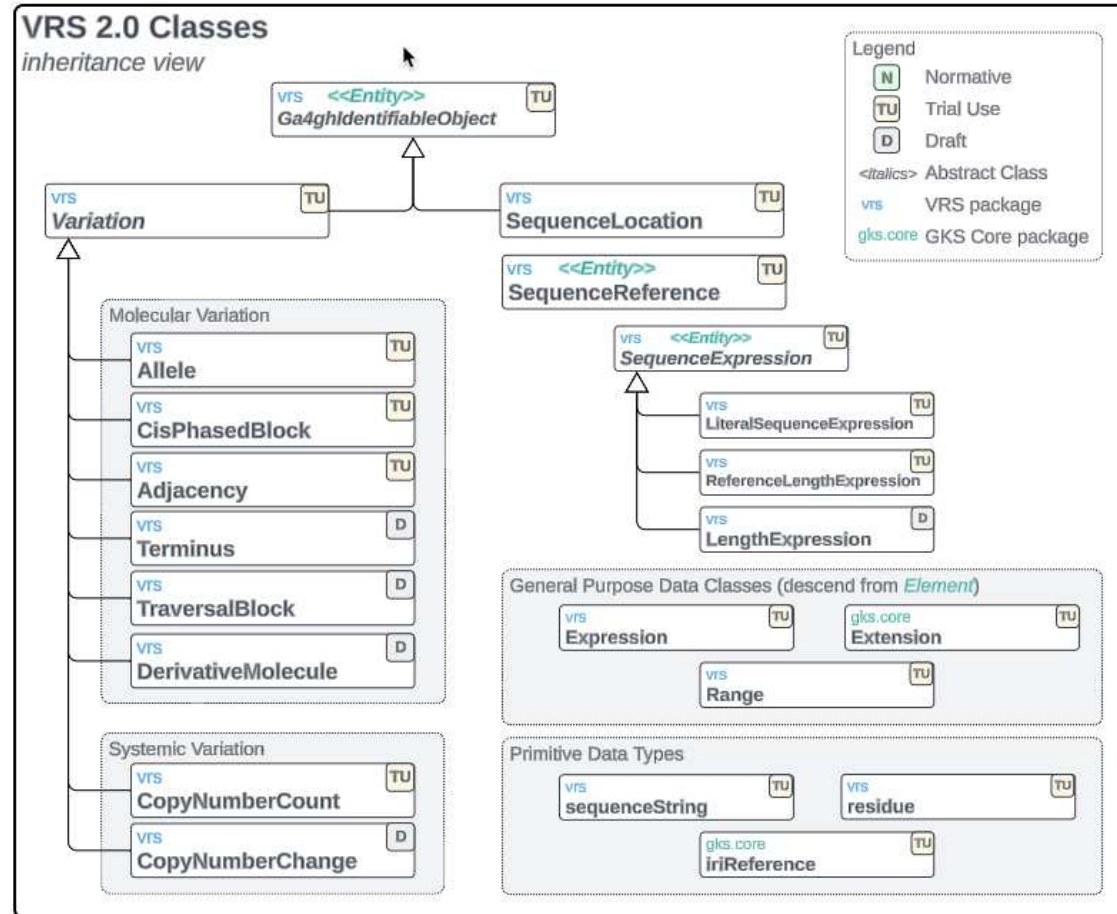
Insertion

Other event

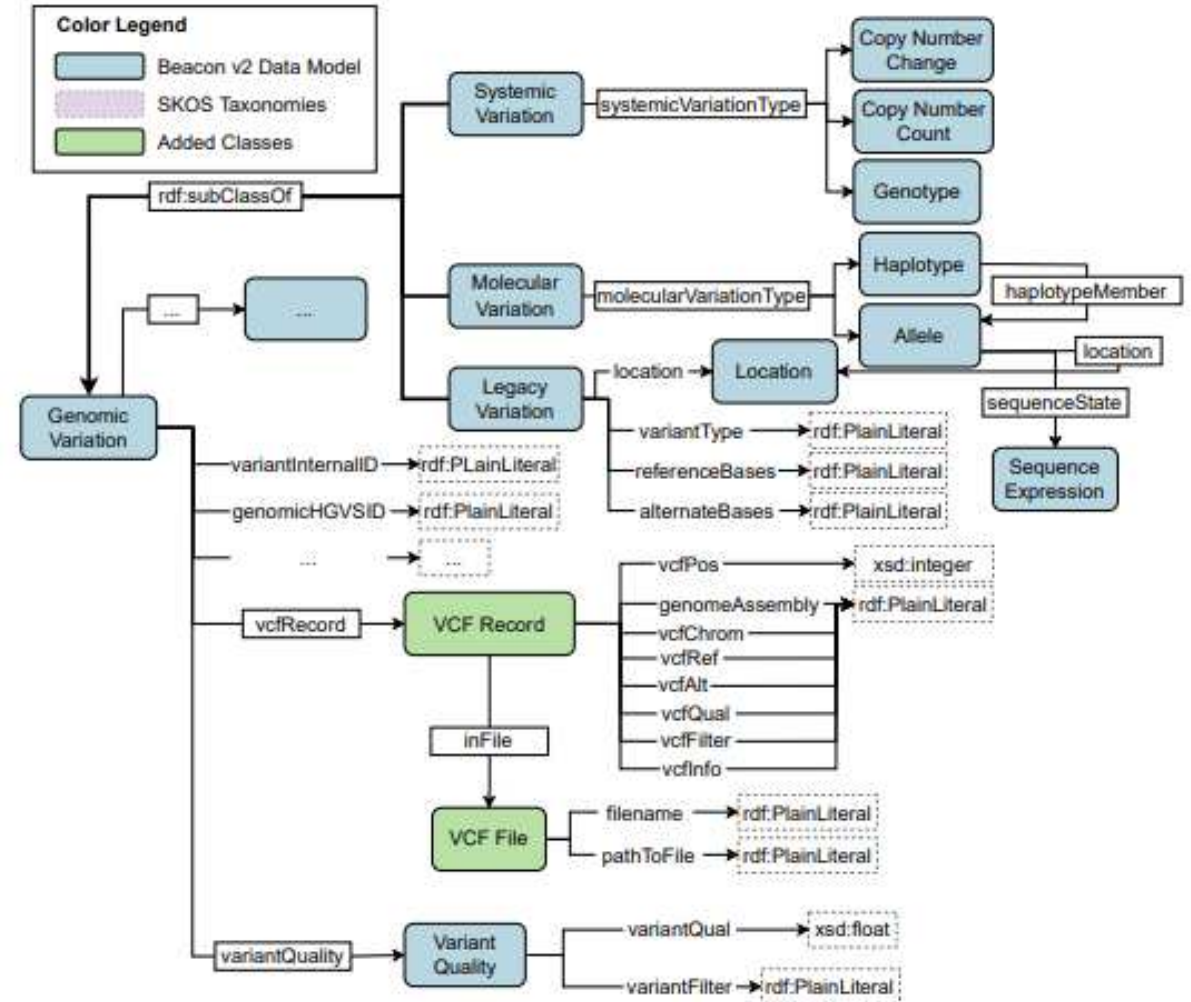
Phased data (G and C above are on the same chromosome)

Representing VCF Semantically?

GA4GH Variant Schema:



HERO Genomics Ontology:

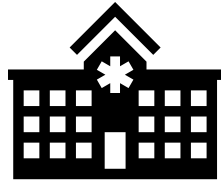
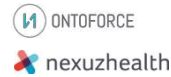


Envisioned Product Evolution



PhD End

PhD
Deliverable



Short Term

Start-up / Product
development
+
Hospital implementation



Medical record



Medium Term

Integrate with EHR
+
Increase application
domains



Long Term

Increase 3rd party
application support
+
Scale to Europe and
beyond